CREATURE ARCHIVE

Field Research Assistant

USER MANUAL

AI-Powered Zoological Field Research Application

Version 1.0 | 2025

For Zoology Students and Field Researchers

1 | Introduction

1.1 About Creature Archive

Creature Archive is an AI-powered mobile application designed specifically for academic zoological field research. It allows researchers and students to capture or upload images of animals, automatically identify species using on-device machine learning, and maintain a structured digital journal of observations — all without requiring an internet connection.

The application bridges the gap between advanced AI species recognition and the practical constraints of remote field work, where internet connectivity is unreliable or completely unavailable.

1.2 Key Features

AI Species ID	Identify animal species from photos with confidence scoring, fully offline
Digital Journal	Create, edit, and manage structured zoological observation records
My Archive	Searchable and filterable library of all past observations
Statistics	Visual analytics including species distribution and 6-month activity trends
Profile & Sync	User account management with cloud synchronisation when online
Dark Mode	Full dark theme support for low-light field environments
Offline First	All core features work without any internet connection

1.3 Intended Users

- Zoology students conducting field research assignments
- Academic researchers studying wildlife in remote locations
- Field biologists requiring structured species documentation
- Any researcher needing offline-capable animal identification tools

ℹ INFO: Offline-First Design

Creature Archive is built for environments with no connectivity. You can identify species, create journal entries, and access your full archive with no internet. Data syncs automatically to the cloud when a connection is available.

1.4 Technical Requirements

Platform	Android / iOS (React Native 0.83+)
Storage	Minimum 200 MB free device storage recommended
Camera	Required for live photo capture (not required for gallery upload)
Internet	Optional — required only for initial account creation and cloud sync
AI Inference	Fully on-device — no server calls required for identification

⚠ WARNING: Storage Warning

The AI model is stored locally on your device and requires approximately 50–100 MB of storage. Ensure your device has sufficient free space before installing. Low storage may cause the app to crash or behave unexpectedly.

2 | Getting Started

2.1 First Launch

When you open Creature Archive for the first time, the app will load the AI model into memory. This may take a few seconds. You will then be presented with the Sign In / Sign Up screen.

> **ℹ INFO: First Load Time**
> The first launch may take 3–5 seconds while the on-device AI model initialises. Subsequent launches are significantly faster. Do not close the app during this process.

2.2 Creating a New Account

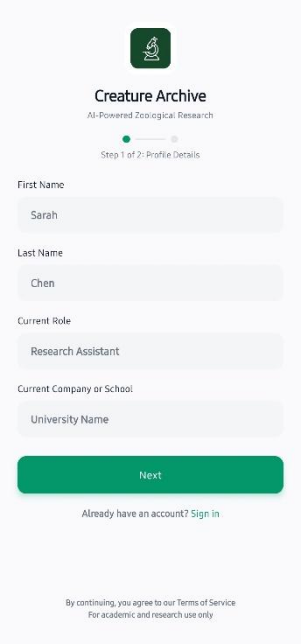
You must create an account to use Creature Archive. An internet connection is required for initial account registration.

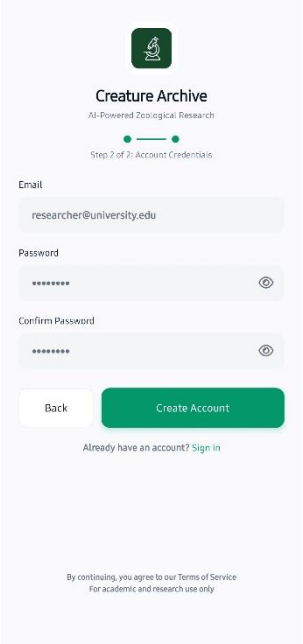
1	On the launch screen, tap 'Sign Up' to begin registration.
2	Enter your First Name and Last Name in the provided fields.
3	Enter your Current Role (e.g., 'PhD Researcher', 'MSc Student').
4	Enter your Current School or Institution name.
5	Tap 'Next' to proceed to the email and password step.
6	Enter a valid email address that you have access to.
7	Create a secure password (minimum 6 characters).
8	Tap 'Create Account'. You will be redirected to the Home Screen on success.

> **⚠ WARNING: Password Requirements**
> Your password must be at least 6 characters long. Passwords are securely handled by Firebase Authentication and are never stored in the database. Choose a strong password combining letters, numbers, and special characters.

✓ TIP: Account Details

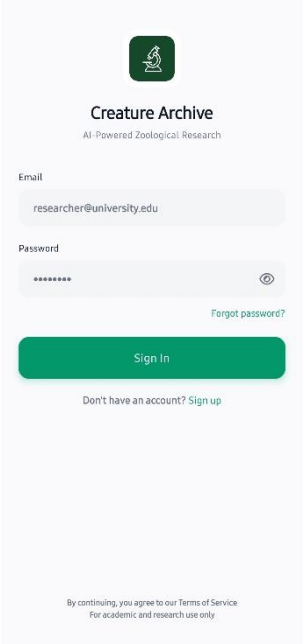
Use your official institution email address. This makes it easier to recover your account if you forget your password and helps maintain data integrity for academic records.





2.3 Signing In

1	Open Creature Archive. The Sign In screen appears by default.
2	Enter your registered email address.
3	Enter your password.
4	Tap 'Login'. You will be taken to the Home Screen with your data loaded.

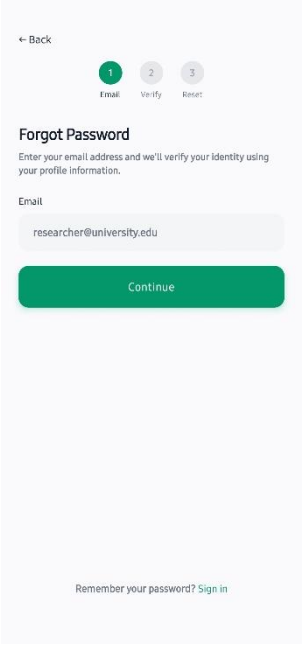


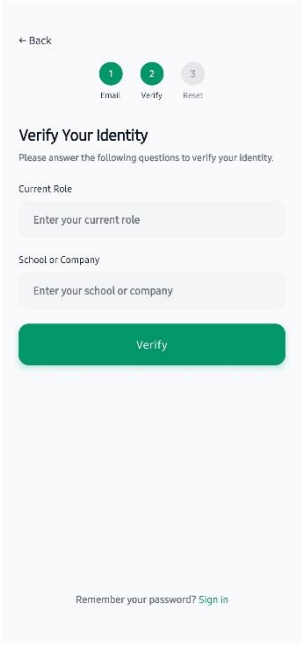
2.4 Forgot Password

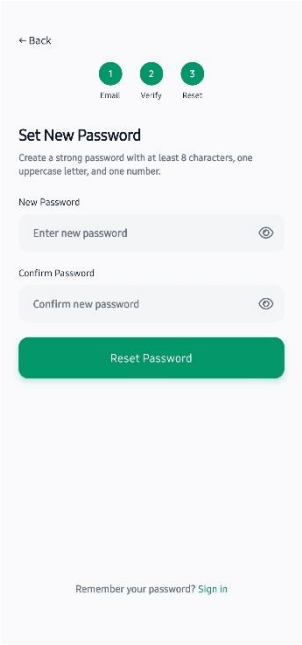
1	On the Sign In screen, tap 'Forgot Password?'
2	Enter your registered email address.
3	Tap 'Send Reset Email'. A password reset link will be sent to your inbox.
4	Open the email and follow the link to set a new password.
5	Return to Creature Archive and sign in with your new password.

⚠ WARNING: Internet Required

Password reset requires an active internet connection. If you are in the field without connectivity, you will not be able to reset your password until you are online. Ensure you remember your credentials before going offline.







3 | Home Screen

3.1 Overview

The Home Screen is the central hub of Creature Archive. It gives you a real-time research summary, quick access to core features, and a live feed of your most recent observations.

3.2 Screen Elements

App Header	Displays 'Creature Archive' title, subtitle 'Field Research Assistant', and your profile avatar (initials or photo) in the top-right corner.
Quick Actions	Three shortcut buttons: 'New Journal' (primary), 'View Archive', and 'Analytics'.
Research Summary	Cards showing Total Observations, This Month count, This Week count, and Today's Observations with a motivational prompt.
Recent Activity	A scrollable list of your most recently created journal entries with species name, scientific name, confidence score, and timestamp.
Bottom Navigation	Persistent tab bar with Home, Archive, Stats, and Profile tabs.

3.3 Quick Actions

New Journal (Primary Button)

The large green 'New Journal' button is the fastest way to record a new observation. Tapping it takes you directly to the Scan / Identification screen where you can capture or upload a species photo.

View Archive

Opens the My Archive screen, showing your complete collection of past journal entries with search and filter capabilities.

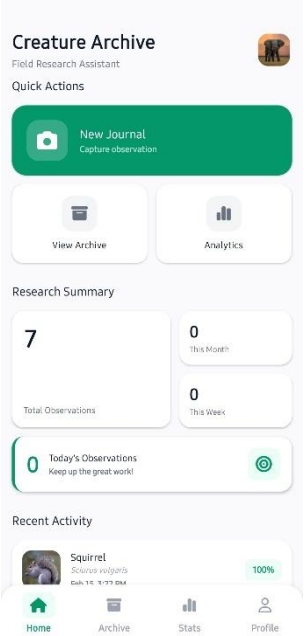
Analytics

Navigates to the Stats screen, where you can view species distribution charts and your 6-month observation activity trend.

3.4 Research Summary Cards

The summary section provides at-a-glance statistics for your research activity. These counters update automatically as you add, edit, or delete journal entries. They reflect your personal data only — no other researchers' data is visible.

> **ℹ INFO: Data Isolation**
> All data shown on the Home Screen belongs exclusively to your account. Creature Archive enforces strict user data isolation — other researchers cannot see your observations, and you cannot see theirs.



4 | AI Species Identification

4.1 Overview

The species identification feature is the core scientific capability of Creature Archive. It uses a MobileNetV2 Convolutional Neural Network (CNN) model trained on thousands of animal species images. The model runs entirely on your device — no image is sent to any server during identification.

> **⚠ WARNING: AI Accuracy Limitation**
> The AI model provides a confidence-scored prediction, not a guaranteed identification. Always verify results against authoritative zoological references, especially for rare, endangered, or closely related species. Never rely solely on AI identification for critical academic or conservation decisions.

4.2 Accessing the Scan Screen

There are two ways to reach the identification screen:

- Tap the green 'New Journal' button on the Home Screen.
- Navigate to Archive and tap the '+' floating action button.

4.3 Capturing or Uploading an Image

Option A: Capture a Photo

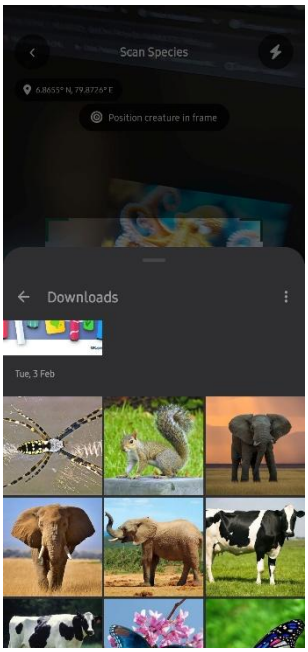
1	On the Scan screen, tap the Camera button.
2	Point your device camera at the animal. Ensure good lighting and that the subject is clearly in frame.
3	Tap the shutter button to capture the image.
4	Review the captured image. Tap 'Retake' to capture again, or 'Use Photo' to proceed to identification.

Option B: Upload from Gallery

1	On the Scan screen, tap the Gallery / Upload button.
2	Your device's photo library will open. Navigate to and select the image you wish to identify.
3	The selected image loads into the Image Review screen for confirmation.

✓ TIP: Image Quality Tips

For best identification accuracy: ensure the animal occupies most of the frame, use natural daylight where possible, keep the camera steady to avoid blur, and avoid heavily backlit scenes. Images are automatically resized and normalised before inference — you do not need to pre-process them.



4.4 Image Review Screen

After capturing or selecting an image, the Image Review screen displays your chosen photo. From here you can:

- Tap 'Identify Species' to begin AI inference.
- Tap back to return and select a different image.

The image is automatically pre-processed (resized, filtered, and normalised) before being passed to the model. This happens on-device and typically takes less than 2 seconds.

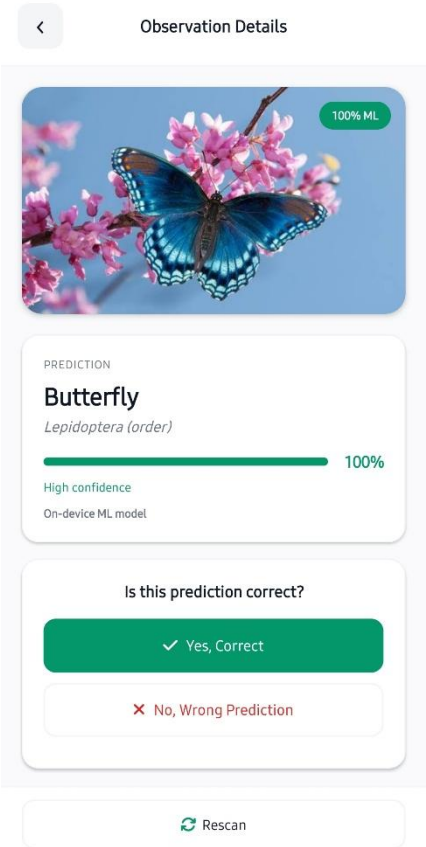
4.5 Viewing Identification Results

Once inference is complete, the Scan screen displays the identification results:

Common Name	The standard English common name of the predicted species (e.g., 'Eastern Gray Squirrel').
Scientific Name	The Linnaean binomial name in italics (e.g., 'Sciurus carolinensis').
Confidence Score	A percentage expressing the model's certainty (e.g., '87%'). Higher is more reliable.
Detection Method	Labelled as 'On-Device AI' or 'Online Detection' based on inference pathway used.

⚠ WARNING: Low Confidence Results

If the confidence score is below 60%, treat the identification with caution. The image may be unclear, the animal may be partially obscured, or the species may be underrepresented in the training data. Consider retaking the photo in better conditions or cross-referencing with field guides.



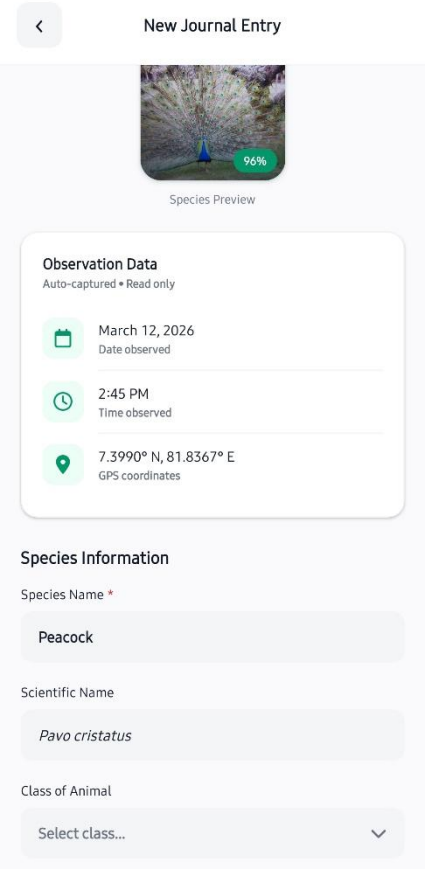
4.6 Creating a Journal Entry from Identification

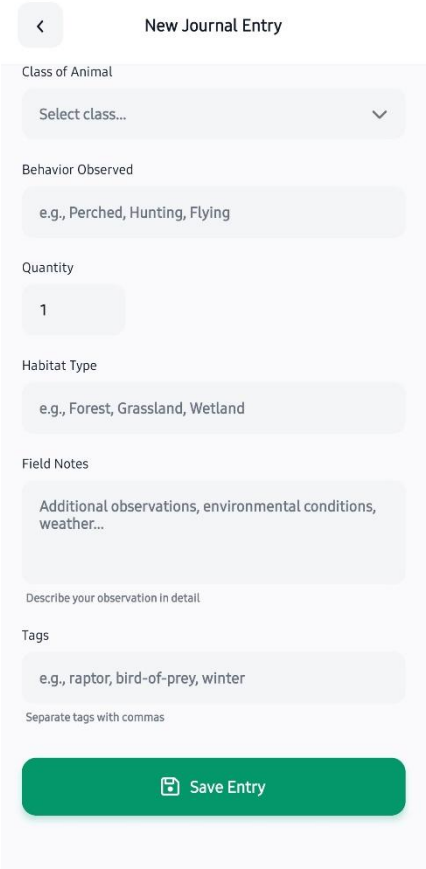
After a successful identification, you are prompted to create a journal entry pre-populated with the identified species data:

1	Review the identification result displayed on screen.
2	Tap 'Yes, Continue' to create a journal entry with the species data pre-filled.
3	Add any additional notes, behavioural observations, habitat details, or tags.
4	Tap 'Save' to store the entry in your archive.

✓ TIP: Pre-populated Data

Species Name, Scientific Name, and Confidence Score are automatically inserted into the new journal entry from the AI result. You can edit these fields manually before saving if needed.

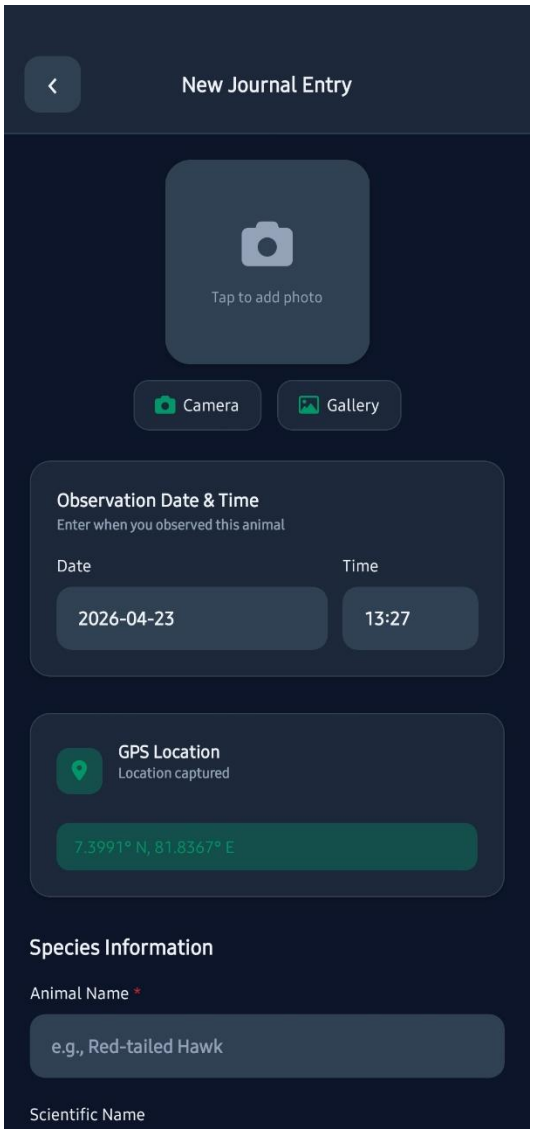




4.7 Manual Entry

You can also create a journal entry entirely manually without using the AI identification:

1	Navigate to My Archive.
2	Tap the '+' Floating Action Button.
3	Select an image from your gallery (or skip if no image is available).
4	Fill in all fields manually: Species Name, Scientific Name, Class, Behaviour, Quantity, Habitat, Notes, and Tags.
5	Tap 'Save'. The entry will be tagged as 'Manual Entry' in your archive.



5 | Managing Journal Entries

5.1 Journal Entry Fields

Each journal entry supports the following fields:

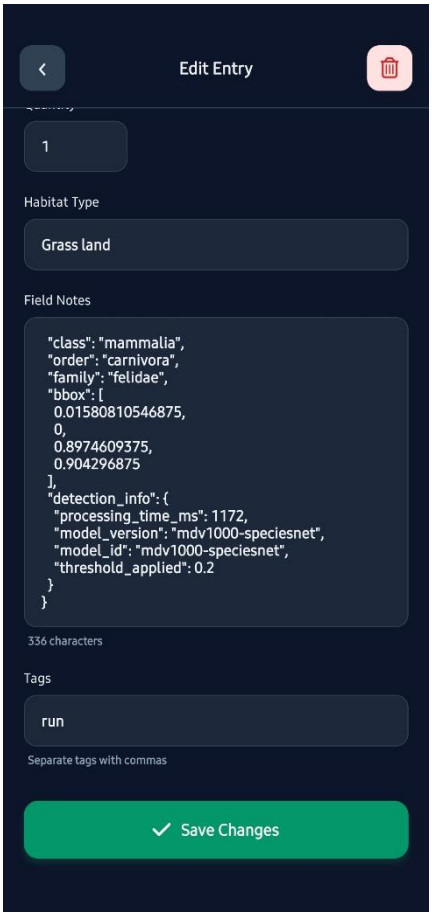
Species Name	Common name of the observed animal (pre-filled from AI or entered manually).
Scientific Name	Binomial nomenclature of the species.
Class of Animal	Taxonomic class (e.g., Mammalia, Aves, Reptilia, Amphibia, Pisces).
Behaviour Observed	A brief description of what the animal was doing at the time of observation.
Quantity	Number of individuals observed.
Habitat	Description of the environment where the animal was observed.
Field Notes	Free-form notes about the observation, conditions, or relevant context.
Tags	Comma-separated keywords for filtering (e.g., 'nocturnal, wetland, rare').
Image	A photo associated with the observation.
Date/Time	Automatically recorded at the moment of entry creation.

5.2 Editing an Existing Entry

1	Navigate to My Archive.
2	Locate the entry you wish to edit. Use search or filters if needed.
3	Tap the entry to open the Observation Detail screen.
4	Tap the Edit (pencil) icon.
5	Modify any fields as required.
6	Tap 'Save Changes'. A confirmation toast appears: 'Entry Updated'.

ℹ INFO: Offline Edits

Edits made while offline are saved locally to your device immediately. A sync indicator will appear on entries that have pending cloud sync. These changes upload automatically when connectivity is restored.

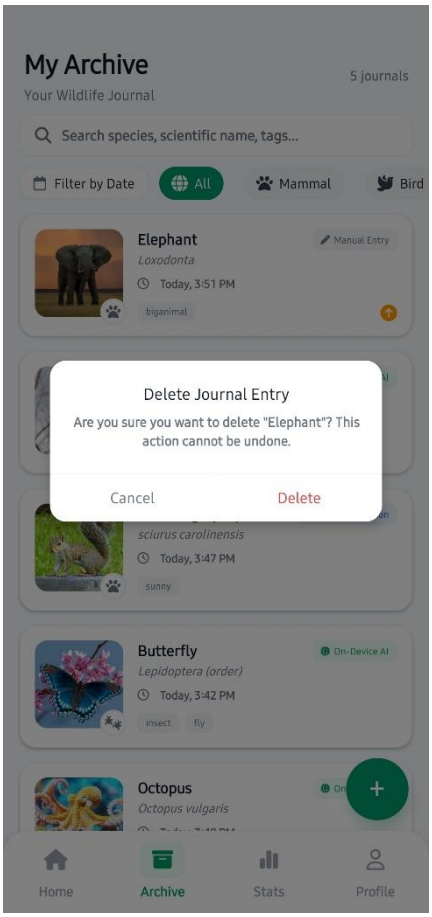


5.3 Deleting an Entry

1	Open the entry from My Archive.
2	Tap the Delete (trash bin) icon in the top-right corner of the Edit Entry screen.
3	Confirm the deletion in the prompt that appears.

✕ CAUTION: Deletion is Permanent

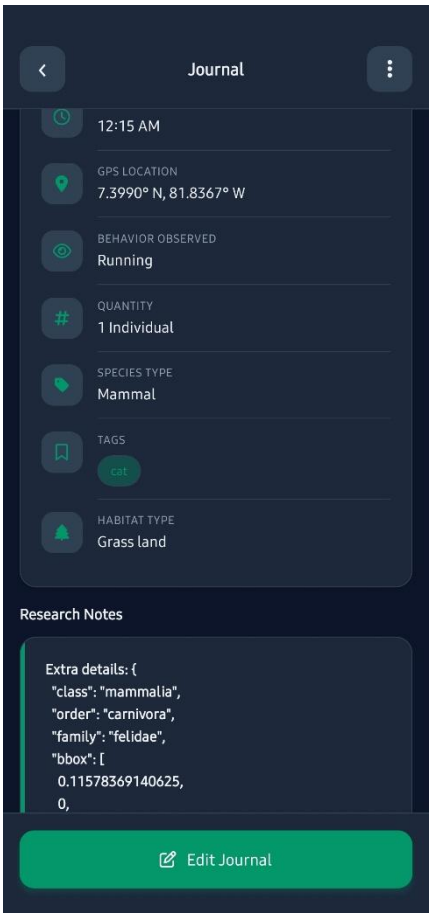
Deleting a journal entry removes it from both local storage and cloud Firestore. This action cannot be undone. Ensure you have a backup or are certain before confirming deletion. Deleted entries cannot be recovered.



5.4 Viewing Observation Detail

Tapping any entry in My Archive opens the Observation Detail screen, which displays the full record including:

- The observation photograph
- Species common and scientific names
- Confidence score and detection method (AI or Manual)
- Taxonomic class, behaviour, quantity, and habitat
- Field notes in full
- Tags and observation timestamp



6 | My Archive

6.1 Overview

My Archive is your complete digital zoological journal — a searchable, filterable library of every observation you have recorded. Access it from the Home Screen Quick Actions, or via the Archive tab in the bottom navigation bar.

6.2 Browsing Your Archive

Journal entries are displayed as cards in reverse-chronological order (most recent first). Each card shows:

- A thumbnail of the observation photo
- Common name and scientific name
- Timestamp of the observation
- Tags associated with the entry
- A badge indicating AI (On-Device AI / Online Detection) or Manual Entry source

6.3 Searching Entries

1	Tap the search bar at the top of the Archive screen.
2	Type any keyword — the app searches species names, scientific names, and tags in real time.
3	Results update as you type. Clear the search field to return to all entries.

6.4 Filtering by Class

Use the filter chips below the search bar to narrow entries by taxonomic class:

- All — shows every entry regardless of class.
- Mammal — filters to entries classified as Mammalia.
- Bird — filters to Aves entries.
- Additional class filters appear as you add entries of different classes.

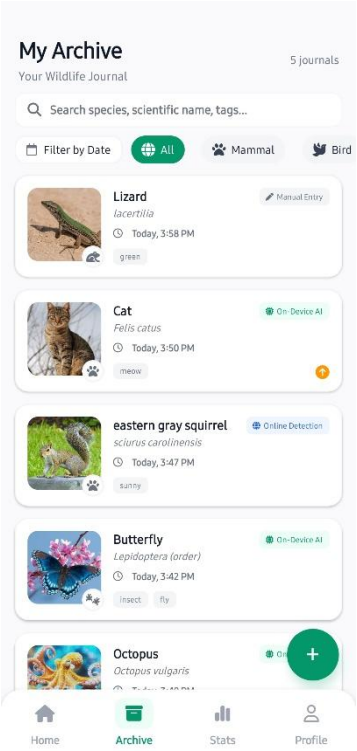
6.5 Filtering by Date

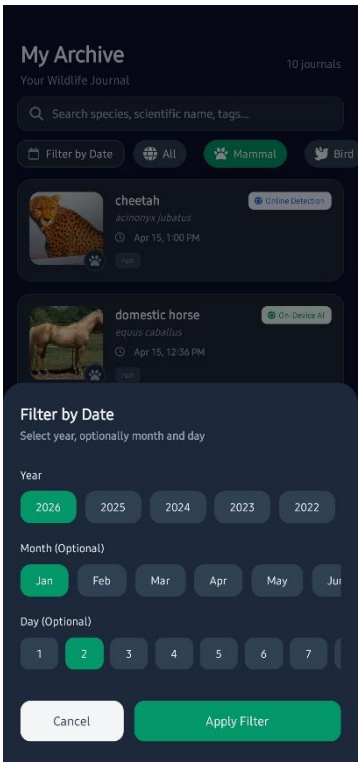
1	Tap the 'Filter by Date' option on the Archive screen.

2	Select a date range using the date picker.
3	The archive will display only entries recorded within the selected period.

✓ TIP: Combining Filters

You can combine text search, class filter, and date filter simultaneously. This is useful when reviewing observations of a specific species class during a particular field expedition date range.





7 | Statistics & Analytics

7.1 Overview

The Stats screen provides visual analytics of your research activity. Access it via the 'Analytics' quick action on the Home Screen or the Stats tab in the bottom navigation bar.

7.2 Species Distribution Chart (Donut Chart)

The donut chart provides a proportional breakdown of your observations by taxonomic class. Each segment represents a different animal class (e.g., Mammals, Birds, Reptiles). Hover or tap a segment to see the exact count and percentage for that class.

> **ℹ INFO: Chart Data Accuracy**
> The chart reflects all entries currently in your archive. If you delete entries, the chart will update accordingly. The chart requires at least one journal entry to render.

7.3 6-Month Activity Trend (Dot Chart)

The trend chart displays your observation activity over the last six months, plotted as a dot-based timeline. Each dot represents an observation, and you can visualise activity peaks and gaps in your research schedule.

Use this chart to:

- Identify productive research periods
- Track whether observation frequency is increasing or declining
- Spot months with no recorded activity that may require fieldwork scheduling

8 | Profile & Settings

8.1 Accessing Your Profile

Tap the Profile tab in the bottom navigation bar, or tap your profile avatar in the top-right corner of the Home Screen, to access the Profile screen.

8.2 Profile Information

Profile Photo	Upload a personal photo from your gallery. Displayed as a circular avatar. If no photo is set, your initials are displayed instead.
Display Name	Your first and last name as registered.
Role	Your academic or professional role.
Institution	Your registered school or institution.

8.3 Uploading a Profile Photo

1	On the Profile screen, tap your avatar or the camera icon.
2	Select an image from your device's photo gallery.
3	The image is uploaded to cloud storage and your avatar updates immediately.

✓ TIP: Initials Fallback

If you prefer not to upload a photo, your initials (e.g., 'FL' for First Last) are displayed as a coloured avatar. This is generated automatically from your registered name.

8.4 Dark Mode

1	On the Profile screen, locate the 'Dark Mode' toggle.
2	Tap the toggle to switch between light and dark themes.
3	Tap 'Save' to persist the preference.
4	The dark theme applies immediately and consistently across all screens.

ℹ INFO: Dark Mode Persistence

Your dark mode preference is saved to local device storage and persists across sessions, even if the app is closed and reopened. The preference also syncs to your cloud profile when online.

8.5 Cloud Sync Toggle

The Cloud Sync toggle controls whether your data synchronises to Firebase Firestore when an internet connection is available:

- Sync ON: Journal entries, edits, and deletions are automatically mirrored to the cloud. This enables cross-device access and provides a backup of your research data.
- Sync OFF: Data is stored only on your local device. No data leaves your device.

⚠ WARNING: Sync Disabled Risk

If Cloud Sync is disabled and your device is lost, damaged, or reset, your journal data cannot be recovered. It is strongly recommended to keep Cloud Sync enabled whenever connectivity is available.

8.6 Signing Out

1	On the Profile screen, scroll to the bottom.
2	Tap the 'Sign Out' button.
3	Confirm sign out if prompted.
4	You will be returned to the Sign In screen. All locally cached data remains on the device.

9 | Offline Use & Data Synchronisation

9.1 How Offline Mode Works

Creature Archive is designed from the ground up to work without internet connectivity. All core functionality — AI identification, journal creation, archive browsing, and statistics — operates using data stored on your device via AsyncStorage.

> **ℹ INFO: Offline Capabilities**
> When offline you can: identify species using on-device AI, create new journal entries, edit and delete existing entries, view your full archive with search and filter, and view your statistics. The only features requiring connectivity are account creation, password reset, profile photo upload, and cloud sync.

9.2 Offline Indicators

Entries that were created or edited offline display an offline indicator badge on their archive card. This confirms the data is queued for sync and has not yet been pushed to the cloud.

9.3 Conflict Resolution

If the same entry is edited on two different devices while one or both are offline, the sync system uses a last-write-wins strategy. The most recent change (by timestamp) is kept. Older conflicting versions are overwritten.

> **⚠ WARNING: Multi-Device Editing Conflicts**
> Avoid editing the same journal entry on multiple devices simultaneously when operating in offline mode. Conflicts will be resolved automatically, but one set of changes will be lost. As a best practice, complete all edits on one device and sync before switching to another.

9.4 Sync Behaviour

When connectivity is restored and Cloud Sync is enabled, the app automatically:

- Uploads all locally created entries that are not yet in the cloud.
- Pushes all locally edited entries to Firestore, overwriting the cloud version.
- Propagates any local deletions to the cloud, removing the entry from Firestore.

✓ TIP: Background Sync

Sync occurs automatically in the background as soon as connectivity is detected. You do not need to manually trigger it. A sync indicator briefly appears during the process.

10 | Troubleshooting

10.1 AI Identification Issues

Problem	The AI returns a low-confidence result (<60%)
Likely Cause	Poor image quality, insufficient lighting, subject too small or partially obscured.
Solution	Retake the photo in natural daylight with the animal clearly in frame. Ensure the animal occupies at least 50% of the image.

Problem	Identification takes longer than expected
Likely Cause	Device has low available memory or is running many background apps.
Solution	Close other applications to free RAM. Restart the app if the issue persists. The AI model targets sub-2-second inference on modern hardware.

Problem	The AI model fails to load on first launch
Likely Cause	Insufficient device storage or memory.
Solution	Free at least 200 MB of device storage and restart the application.

10.2 Authentication Issues

Problem	Cannot create an account
Likely Cause	No internet connection, or email address already in use.
Solution	Check connectivity. If the email is already registered, use 'Forgot Password?' to recover access.

Problem	Incorrect password error at sign-in
Likely Cause	Wrong password entered, or account does not exist.
Solution	Use 'Forgot Password?' to reset via email. Ensure you are using the correct email address associated with your account.

10.3 Sync & Data Issues

Problem	Entries created offline are not appearing after reconnecting
Likely Cause	Sync may still be in progress, or Cloud Sync is disabled.
Solution	Check that Cloud Sync is toggled ON in the Profile screen. Wait a few seconds after restoring connectivity for background sync to complete.

Problem	Journal entry data appears different on another device
Likely Cause	Unsynchronised edits from one of the devices; conflict resolved by last-write-wins.
Solution	Always ensure you sync one device fully before editing on another. Re-enter lost data manually if necessary.

10.4 App Performance Issues

Problem	App is slow to launch
Likely Cause	AI model initialisation on first run, or low device resources.
Solution	Allow 5–10 seconds on first launch. Subsequent launches will be faster.

Problem	Archive list scrolls slowly with many entries
Likely Cause	Large number of high-resolution images in the archive.
Solution	This is expected behaviour with very large archives. Performance is optimised for typical research use volumes (hundreds of entries). Extremely large archives (thousands of entries) may experience degraded scrolling.

11 | Data Privacy & Security

11.1 Data Storage

Local Storage	Journal entries and app preferences are stored on-device using AsyncStorage. This data is accessible only by the Creature Archive app.
Cloud Storage	When Cloud Sync is enabled, journal entries are stored in Firebase Firestore. Your account data is stored in Firebase Authentication.
AI Inference	All species identification happens on-device. Your images are never transmitted to any server for AI processing.
Profile Photos	Profile images are uploaded to Firebase Cloud Storage and linked to your account.

11.2 User Data Isolation

Creature Archive enforces strict user data isolation at the database level using Firebase Security Rules. Your journal entries are stored under your unique user ID and can only be read or written by your authenticated session. No other user, researcher, or administrator can access your observation data.

ℹ INFO: Security Architecture

Passwords are handled exclusively by Firebase Authentication. No password data is stored in the Firestore database. Authentication tokens expire and are refreshed securely. Your credentials are never transmitted in plain text.

11.3 Responsible Use

✕ CAUTION: Species Data Responsibility

If your research involves endangered, rare, or legally protected species, be aware that detailed location-stamped observation records may have conservation and legal implications. Consult your institution's ethical research guidelines before recording and storing precise location data for sensitive species observations.

Creature Archive is designed for academic use. Users are responsible for ensuring their use of the application complies with their institution's ethics policies, local wildlife observation laws, and data protection regulations applicable in their jurisdiction.

12 | Glossary

AI / Artificial Intelligence	The use of machine learning algorithms to perform tasks that typically require human intelligence, such as species identification from images.
AsyncStorage	A local key-value storage system used by the app to persist data on your device without an internet connection.
Binomial Nomenclature	The formal two-part Latin naming system for species (e.g., Felis catus for domestic cat).
CNN (Convolutional Neural Network)	The type of deep learning model used by Creature Archive for species classification from images.
Cloud Sync	The process of automatically uploading and synchronising local device data to Firebase Firestore over the internet.
Confidence Score	A percentage produced by the AI model expressing how certain it is about its species identification prediction.
FAB (Floating Action Button)	The '+' circular button on screen used to create new journal entries.
Firebase	Google's cloud platform used for user authentication, database storage (Firestore), and cloud file storage.
Firestore	Firebase's NoSQL cloud database where journal entries and user profile data are stored.
MobileNetV2	A lightweight, efficient deep learning architecture optimised for on-device inference on mobile hardware.
On-Device AI	AI inference that runs entirely on the user's device, without sending data to any external server.
TensorFlow Lite (TFLite)	A lightweight version of TensorFlow used to run the trained species identification model on mobile devices.
Taxonomic Class	A level in biological classification (e.g., Mammalia, Aves, Reptilia) used to group related species.

13 | Quick Reference Card

13.1 Navigation Summary

Tab / Button	Function
Home Tab	Summary dashboard, quick actions, recent observations
Archive Tab	Full journal library with search and filters
Stats Tab	Species distribution donut chart and 6-month activity trend
Profile Tab	Account settings, dark mode, cloud sync, sign out
New Journal	Start new observation (camera or gallery)
+ FAB (Archive)	Create new manual journal entry
Entry Card Tap	Open Observation Detail screen
Edit Icon	Edit an existing journal entry
Delete Icon	Permanently delete a journal entry

13.2 Warning Summary

Warning	Description
Deletion is permanent	Deleted entries cannot be recovered. No undo.
AI accuracy limitation	AI provides predictions only. Verify with authoritative sources.
Low confidence (<60%)	Retake photo in better conditions before trusting result.
Password required for reset	Internet needed to reset password. Remember credentials offline.
Sync disabled data loss	Disable sync at risk of losing data if device is lost or reset.
Multi-device conflict	Editing offline on two devices may cause data loss due to conflict resolution.
Storage requirement	App requires ~200 MB free storage for the AI model.

Creature Archive

Field Research Assistant | User Manual v1.0

Built for academic researchers in the field.

React Native | TensorFlow Lite | Firebase | MobileNetV2 CNN